

Battery Management Systems (BMS) for Increasing Battery Life Time

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Abstract

Reliability and costs of an energy storage system are two very important parameters for uninterruptable power supplies (UPS) and other battery applications. The increasing of battery life and the prediction of battery failure are therefore two important features of a battery management system (BMS). A couple of other helpful features can be implemented in a BMS. In case of advanced batteries special features like thermal management (i.e. high temperature batteries) or safety management (i.e. Lithium batteries) is necessary. Therefore the BMS must be optimized to the battery type and the application. In smaller systems the BMS is connected very close to the battery or inside the battery and an operation without the BMS is not possible.

1 Introduction

For telecommunication applications a power supply with high reliability is necessary. In most cases the application has a back-up power supply based on a battery storage. However electrochemical batteries have a life time that is much shorter than the life time of the other components. To detect battery failures in time, monitoring and battery state determination of the battery are necessary. The battery state is mainly based on the state of charge (SOC) and the state of health (SOH). The knowledge of the battery state gives the possibility to close the control loop by using the battery state for the electrical management (i.e. charge method).

2 Parameters influencing battery lifetime

In general battery ageing parameters can be separated in internal parameters and external parameters. Internal parameters describe all parts that can be influenced by the battery manufacturer (i.e. Grid-alloy, manufacturing quality etc.) whereas external parameters describe the ageing by operation. Typical external parameters are operation temperature, charge method, maintenance, deep discharge protection and the electric cycling regime. A BMS cannot influence the internal parameters and also some external parameters are fixed by the type of application. However there are some external parameters than can be controlled in a more or less wide range by a BMS (i.e. charge method, deep discharge protection). One target of an advanced BMS is the optimisation of this adjustable parameters as a function of other operation conditions and the battery history.

2.1 Temperature

Especially high temperatures and temperature differences between cells reduce battery life time. Therefore the design of the whole system (closed cabinet or free standing) have an influence to the life time. In case of high current applications and closed cabinet construction a thermal management system can be used to extend battery life time.

2.2 Charge method

It is well known that charging influences battery lifetime significantly. In case of lead acid batteries overcharging and insufficient charging will cause an reduction in life time. The optimum charge method depends on internal battery parameter, battery temperature and operation history.

2.3 Deep discharge protection

Deep discharge, especially long rest periods at low SOC reduces battery lifetime. In most cases deep discharge protection is based only on the battery voltage (sometimes on the current too). However, by this methods it is not possible to detect critical operation situations in case of small discharge currents. Figure 1 shows the relation between battery voltage, discharge current and DOD at room temperature for a pasted plate battery. The figure shows that a simple voltage threshold will cause very different DOD at the cut off point. Especially in case of small discharge currents the battery will be over-discharged, resulting in damaging the battery.

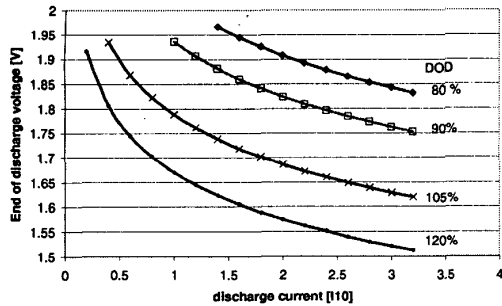


Figure 1: Battery voltage vs. discharge current in relation to the DOD.

2.4 Influence of operation mode

The kind of operation (electrical cycling regime) influences battery life time too. The following figure shows the capacity of three identical batteries (lead acid, vented type, tubular plate) at different test regimes.

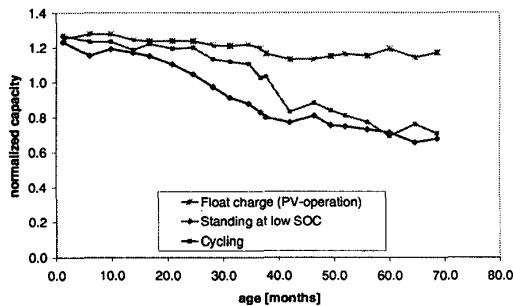


Figure 2: Capacity vs. life time of a LAB for different operation modes
reference battery: float charge (2.25V/cell)
cycling battery: PV related cycling
standing battery: Rest at low SOC (30%)

3 Battery Management

3.1 General

To increase the life time of the battery the external battery parameter are to manage. This is not only important for the lead-acid battery but also for the advanced battery systems like lithium batteries and high temperature batteries. Therefore, a battery management system (BMS) is required taking into account the special demands of the battery technology.

As input signals voltages, currents and temperatures are measured. The function of a battery management system can be split into the following tasks:

- Data acquisition
- Battery state determination
- Electrical management
- Thermal management (not always necessary)
- Safety management
- Communication

Figure 3 shows a simplified schematic drawing of a BMS.

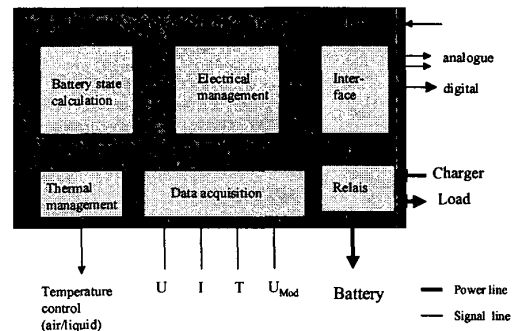


Figure 3: Schematic structure of a battery management system (BMS)

In case of small batteries some of the above mentioned functions are available as single or multiple chip solutions. [1]. For example Li-Ion battery packs for cellular phones and laptop computers contain as a minimum a safety management system. In case of larger battery systems, the battery management system is more complex and BMS systems must individually developed for the battery technology and the application[2].

3.2 Data acquisition

All algorithm of the BMS use measured and calculated data as input information. Therefore the accuracy, the sampling rate and the characterisation of front end filtering are very important and depend on the type of application. For example the sampling rates of EV applications are faster than one sample per second whereas in case of photovoltaic or UPS applications sampling rates below 0.2 samples per second are used.

A critical point within UPS applications are single cell/module failures, leading in reduced operation time. Therefore module- or cell-voltage measurements are very important and part of UPS management

systems. Two different architectures are known for module measurements:

- In case of a centralised system the measurement unit is located within the BMS. The advantage of this technology is the lower material costs. The main disadvantage is the wiring between the BMS central unit and the battery cells (modules).
- In case of a decentralised system the measurement unit is decentralised in cell/module oriented units. This units communicate with the central BMS via a digital communication bus. The main advantage is the reduced wiring and the possibility to expand the system. Additionally the decentralised units can measure the temperatures of the cells/modules with a minimum of costs. This additional information gives the possibility for better, but also more complex, battery state calculation. Decentralised systems get more economic in case of larger batteries.

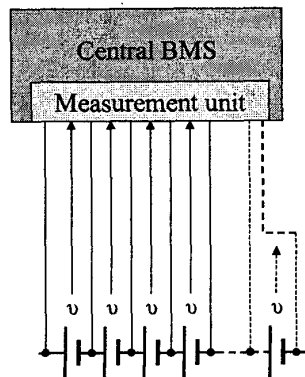


Figure 4: Centralised data acquisition

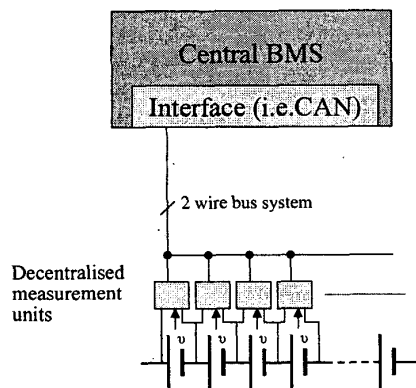


Figure 5: Decentralised data acquisition

3.3 Battery state calculation

The battery state is used as input parameter for the electrical management and additionally it is an important parameter for the user. So the battery state

can be used to estimate the bridging time or the expected lifetime of the battery.

The state of the battery can be simplified described by the following two parameters:

- State of charge (SOC)
- State of health (SOH)

Both parameters are not independent and influence the battery performance (i.e. available capacity). The relationship is shown in Figure 6.

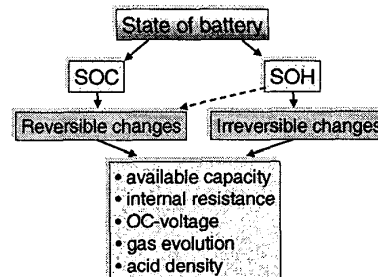


Figure 6: Relation between SOC and SOH

For SOC determination in most cases Ah counting, including charge loss estimation are used[3]. For SOH determination a couple of more or less good working methods are known and used. It depends strongly on the battery technology and the kind of application which method is useable.

Figure 7 shows a method using an optimum filter algorithm for SOC and SOH determination. Other methods like fuzzy clustering and neural networks were also developed and tested [4].

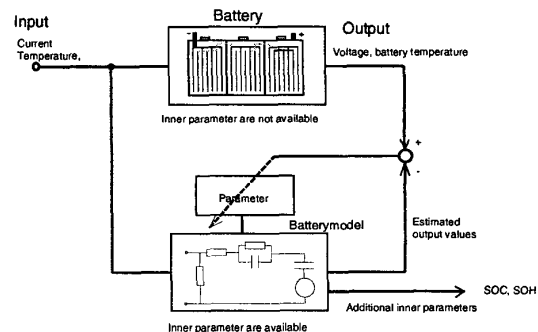


Figure 7: Adaptive method for battery state determination

As optimum filter a Kalman Filter is used. The basic of such a filter is a numerical battery model. The Kalman Filter takes the statistical knowledge of the parameters and the measurement into account. The deterministic knowledge is described by differential equations which are included in a battery model. This model is corrected by statistical knowledge. The main

advantage of this method in comparison to Ah-counting is that the state of charge calculation is not influenced during long term operation (problem of drift).

3.4 Electrical management

Within the electrical management the input parameters current, voltages, temperatures, SOC and SOH are used to control the charge and discharge process. The following tasks must be fulfilled:

- Control of the charge process, including equalization charge and float charge
- Control of the discharge

The charge control process and the discharge limitation depends strongly on the used battery technology and the battery type. For UPS systems the recharging of the battery must be adapted to the deep of the previous discharge, the temperature and optionally to the battery age. Quick recharge in combination with lowest aging are the main targets. Additionally the control of the float charge is very important.

Figure 8 shows the limitation of the discharge current in case of an electric vehicle. In case of UPS applications the discharge can only be controlled by a priority controlled cut off of the loads.

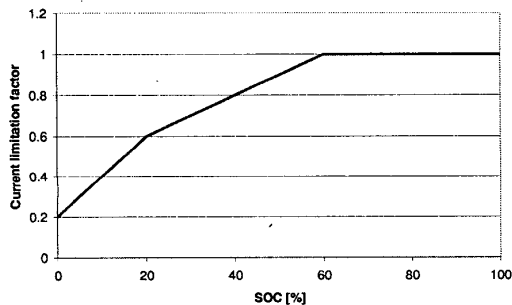


Figure 8: Function for current limitation vs. SOC as it is used in electric vehicles

3.5 Safety and supervisor management

The safety management has to protect the battery against critical operation conditions. In case of an BMS for an electric vehicle, the tasks of the safety management system are:

- Protection against deep discharge
- Protection of single cells against overdischarge
- Protection against overtemperature (thermal runaway)
- Battery turn off in case of a crash

Overcurrent and overvoltage should also be detected by the safety management.

3.6 Thermal management

A thermal management is necessary for most high power applications and for high temperature batteries. The tasks of the thermal management system are temperature equalization between the cells, cooling of the battery and in some cases, i.e. high temperature batteries, the heating of the battery. Therefore liquid or air (fan) systems are used. One major problem is the heat transfer from the inside of the battery cells to the outside, because the thermal resistance of the electrolyte and the plastic cell housing is quite high.

In case UPS systems the equalization between the cell temperatures is the most important function of a thermal management system. In case of a closed unit containing the rectifier, the inverter and the batteries high temperature gradients within the battery are possible. In case of larger UPS systems where the battery is not located together with power electronics in a cabinet, thermal management is less important.

3.7 Communication

The communication between the BMS and other onboard and off-board devices is another important task of the BMS. Depending on the application different interface systems for data exchange are used:

- Analogous signals
- Pulse wide modulated signals
- Serial Interface

Newer UPS systems use serial interfaces like CAN (control area network). This communication interface gets more and more popular for automation systems and automotive use. A lot of hardware components are available supporting this interface standard.

¹ Philips Semiconductors: Data Sheet SAA 1502ATS, Safety IC for Li-ion, Eindhoven, the Netherlands, 1998

² D. Heinemann et al.: A new design of a battery management system including a range forecast, EVS 14, 1997

³ A. Jossen et al.: Battery Control Unit with State of Charge Indicator, 10th European Photovoltaic Solar Energy Conference and Exhibition, 1991, Lisbon

⁴ J. Garche et al.: Failure modes and the detection of the state of health of lead-acid batteries in PV-systems, LABAT, 1996